FANSTEEL INC.:

THE ADVANCED STRUCTURES DIVISION

"If you want to see an example of the truly enormous problem of breathing life into an abstract plan for business success, go and study our Advanced Structures Division," Mr. Warren B. Hayes, the president and chief executive of Fansteel Inc., told the case writer from the Harvard Business School. He continued, "Fred Fletcher, the division manager, is facing issues that are quite tough, both conceptually and administratively. He is trying to weave together the activities of eight separate business firms with an organization that will permit a high degree of autonomy for the operating managers and still allow full advantage to be taken of the opportunities for coordinating their technological and manufacturing capabilities."

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Since its creation in 1914, Fansteel Inc. had been a leading producer of refractory metals[1] and related products. In 1964, Mr. Hayes significantly altered the traditional course of the company by redefining its goal as that of becoming "an integrated supplier of material systems." The key elements of this new corporate concept were the decision to base the future growth of the company upon competence in a broad range of materials technologies, a commitment to serve those markets which were judged to be most sensitive to advances in materials technology, and the decision to accomplish these objectives by forward integration into various fabrication activities and horizontal integration across the new capabilities that were acquired or developed. In 1964, the Advanced Structures Division was formed to act as the primary organizational expression of this strategy.

Growing chiefly through acquisition, by November 1967 the Advanced Structures Division (ASD) consisted of eight separate business entities, four of which had been acquired within the last ten months. These business units,

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[1]The refractory metals are tantalum, tungsten, molybdenum, and columbium. They are valued for their strength at high temperatures, resistance to chemical corrosion, and other special metallurgical and electrical properties.

This case was written in 1967 by Richard P. Rumelt under the supervision of Prof. William Guth at the Harvard Business School. It was never copyright nor used in any teaching capacity.

called the "plants" by members of division management, were all specialty job shops, located in the Los Angeles area, with annual sales ranging from $400,000 to $8.3 million. Operating virtually autonomously, each possessed distinctive skills in the manufacture to customer order of parts and structures for the aerospace, chemical, and nuclear industries. Despite the appellation "plant," these businesses were fully capable of operating on an independent basis, each having its own facilities, labor force, engineers, salesmen, and management group. ASD's management estimated that the division's total sales for 1967 would be slightly in excess of $20 million, in contrast to $9.1 million in 1966.

The rapid expansion of the division had, by November 1967, heightened management's awareness of the problem of organization: Mr. Frederick R. Fletcher, the general manager of ASD and vice president of Fansteel, was studying the problem of developing patterns of responsibility, coordination, and control that would allow the division to act as a more cohesive whole in the market, and thereby more effectively accomplish the over-all mission he had defined as "performing a materials system management function." In particular, Mr. Fletcher had identified four issues that were critical to the future progress of the division:

1. ASD's operations were becoming too varied and complex for Mr. Fletcher to remain in close contact with the plant managers on a day-to-day basis. Consequently, he was considering the advisability of clustering the plants into groups and assigning the responsibility for the performance and development of these larger units to "group managers." He had not as yet decided, however, whether the groupings should be based upon common technologies, common manufacturing operation, similar levels of technical sophistication, or some other criterion.

2. The "materials system management function" was essentially that of maintaining a close control over the entire sequence of manufacturing and engineering activities, from raw materials to finished products, especially when the skills of several different plants were required. Since an increasing proportion of the division's business was expected to require the use of more than one fabrication technology, Mr. Fletcher was convinced that it was important for ASD to be able to achieve a low cost of coordination among the plants. Equally important would be the representation of this coordinated capability to the customer.

3. Much of the future growth of the division, Mr. Fletcher believed, would be due to the penetration of new markets. This would result, he said, from two different types of innovative activity. First, new markets could be opened up through the development of new materials and new processes for fabricating products. Secondly, current know-how in materials and fabrication technology could be brought to bear on product design problems encountered by customers with outdated views about materials limitations. To accomplish these tasks, ASD would have to have a group, or groups, capable of sophisticated technical development work. In addition, engineers in customer firms would have to have a single point of contact within ASD which could provide them with technical problem-solving skills encompassing the whole range of materials fabrication techniques. At present, no such groups existed at ASD.

4. The approach taken toward coordinating the division's efforts would have to complement a policy of plant manager autonomy. The abilities of the managers of the individual plants, Mr. Fletcher maintained, represented a large portion of the value of those businesses. "The plant managers," he said, "must be able to compete effectively in their own markets on an independent basis. The materials fabrication markets are quite competitive and require close management attention to detail. It would be foolhardy to attempt to integrate the plants by any method that impaired their individual effectiveness. By giving the plant managers full and complete responsibility for their businesses, we can insure that the basic building blocks of the division remain strong and profitable."

Fansteel - An Overview

Fansteel, in 1967, was organized into four operating divisions: Metals, Electrometals, VR/Wesson, and Advanced Structures. The combined net revenues of these divisions in 1966 was $53.7 million, on which the company realized a net income of $2.5 million. The mean net margin for the major producers of refractory metals was 5.5% in 1966, although the performance of individual firms varied widely from this figure. A breakdown of the contribution to net revenues and profitability of each of Fansteel's divisions in 1966 is shown in Table 1.

Table 1

Contribution to Net Revenues and Profitability of Fansteel's Divisions in 1966

Division	Proportion of Net Revenues	Net After-Tax Margin on Sales
Metals	19.6%	5.5%
Electrometals	27.1	6.8
VR/Wesson	35.8	11.7
ASD	17.5	0.7

Management estimated that the company's sales for 1967 would be close to $80 million and that the net margin would be about 5%, based upon figures available in October.

The Metals Division was the hub of the corporate system, producing the refractory metals that served as inputs to the manufacturing operations of the other divisions, as well as selling these metals to other industrial users. It produced high purity tantalum, columbium, tungsten, and molybdenum in a number of forms, including bar, sheet, foil, wire, rod, and powder, as well as special alloys. About 66% of its sales were to the external market. Over one-half of its external sales consisted of high-grade tantalum powder and foil to electronics component manufacturers for use in tantalum capacitors. About 26% of the division's internal sales was tungsten powder to the VR/Wesson division.

The invention that launched Fansteel (then the Pfanstiel Electrical Laboratories) in 1914 was the development of a process for inexpensively cutting tungsten contact disks from tungsten rod, these disks being used for electrical contacts in automobile spark coils. In 1967, tungsten was still the favored material for electrical contacts, and Fansteel's second largest division, Electrometals, derived over 90% of its revenues from their sale. About 40% of Electrometal's output was consumed by the automobile industry and the automotive aftermarket.

VR/Wesson was formed in 1961 when Fansteel acquired the Wesson Tool Company and combined it with its own VR cutting tool division. It was the largest, most profitable, and most autonomous of the divisions. VR/Wesson's major products were super-hard cutting tools made of tungsten carbide and titanium carbide. These cutting tools were used in conjunction with machine tools in the shaping of metal parts. This specialized part of the cutting tool industry was growing at about 20% per year, with the key markets being the automotive industry, farm equipment manufacturers, and other appliance producers. Over 50% of the cash generated within Fansteel was provided by VR/Wesson. Corporate management looked upon the large net cash flow it provided as a vitally important resource for the company, financing growth in all parts of the firm.

Fansteel did not have a central research laboratory, nor was there any group within the company that performed "basic research." Mr. Hayes was a strong believer in the decentralization of research and development: "There are very few successful centralized research groups in the industrial world. It is just too difficult to get the technology out of the laboratory and into the divisions." The formal research and development activities that existed in November 1967 were:

1. The Electronic Materials Laboratory (EML) in the Metals Division. This facility was engaged in developing new applications for the refractory metals in the field of electronic components and in improving the current products of the Metals Division.

2. The Powder Metals Plant, one of the business units of the Advanced Structures Division, was working on new methods for creating metal powders and forming them into parts.

3. San Fernando Laboratories, newly added to the Advanced Structures Division in 1967, specialized in developing applications for "chemical vapor deposition" processes, a technique for forming a metal part on a molecule by molecule basis.

4. The Applied Physical Metallurgy Laboratory, nominally reporting to the division manager of ASD, was engaged in developing the technology of "plasma arc flame spray processes."

While not yet acquired by Fansteel, negotiations for the purchase of the assets, organization, and patents of the Du Pont Metals Center were essentially complete. Management said that this facility would provide the company

with important proprietary technologies and a group of metallurgists and chemists with a proven record of ability. Mr. Hayes planned to add the Du Pont laboratory to the Metals Division, and noted that "I expect that this new acquisition will make a sudden difference as to how research and development is done in Fansteel. One of the things I see happening is a transfer of some of the research type activities that have been building on the West Coast (ASD) to the Metals Division and to Du Pont."

The Development of a New Corporate Concept

In the decade between 1950 and 1960, Fansteel was best known as the world's leading producer of tantalum capacitors,[1] supplying close to 100% of the rapidly growing demand for these devices. Paradoxically, the very success of this product line sowed the seeds of disaster; research and engineering budgets were gradually cut, the best of the firm's technical people left, and the over-all ability of the company to innovate and compete vigorously atrophied. When, in 1960, several electronics component manufacturers introduced superior tantalum capacitors, Fansteel was unable to develop a competitive device. Its market share, once 100%, fell to 2% by 1962.

The Board of Directors decided to seek out an individual who had distinguished himself in the electronics component industry and offer him both the position of president and the challenge of effecting an improvement in the company's position. In February 1963, the Board elected Mr. Warren B. Hayes to the office of president. At 39, Mr. Hayes had previously been the president of TRW Electronics Inc., a subsidiary of Thompson Ramo Wooldridge Inc.

Mr. Hayes immediately inaugurated a cost reduction program, managed to stem the growing losses in the capacitor business, and struggled throughout 1964 to turn the business around. Finally, near the end of 1964, he decided to take a bold step - to eliminate the product line he had been hired to improve. "I decided," he said, "that the goal wasn't worth the price. Fansteel had more attractive opportunities to which it could commit resources."

Search for Opportunity. The disposition of the tantalum capacitor business marked the end of an era for Fansteel and forced management to turn its attention once more to the development of the company's historic competence - the refractory metals.

From a corporate point of view, Fansteel's principal skill lay in the extraction and consolidation[2] of the refractory metals. The company and

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[1] A capacitor is a small, relatively simple device which is used in almost every electronic circuit.

[2] Extraction and consolidation were the steps involved in producing a piece of pure metal from ore. Conversion and fabrication were the subsequent steps in which the metal, or an alloy, was cut, formed, welded, forged, cast, rolled, or otherwise transformed into the final desired shape.

its competitors could be roughly characterized as being in capital intensive, high fixed cost businesses with considerable market uncertainty. These factors created a strong pressure on the firms in the special metals[1] industry to seek out ways of expanding and stabilizing the utilization of their investment in plant and equipment.

The special metals found their chief use as engineering materials that could stand up to harsh environments. It was, therefore, expected that the aerospace, chemical equipment, and nuclear equipment industries would constitute the primary market for these materials as the engineering design problems faced in these fields frequently arose from materials performance requirements in particularly hot, corrosive, radioactive, etc., environments. Since the early 1950's, industry experts had predicted that demand for the special metals would dramatically increase as a result of their utility in these growth industries. The expected increase, however, never appeared. Although the special metals possessed many desirable properties, their use had been limited by the difficulties in fabricating them into products and assemblies. The very properties which made them desirable also made them very hard to work with. The traditional methods of changing the shape of a piece of metal were to apply heat and some form of chemical, electrical or mechanical force: cutting, rolling, casting, etching, welding, forging, etc. The refractory metals were so resistive to deformation and to weakening at high temperature that these methods, for the most part, would not work. Tantalum, for example, when heated to a high enough temperature to soften it, will react violently with the air or with the crucible or holder that contains it.

"In looking at the special metals industry," said Mr. Hayes, "my first impression was that breakthroughs in fabrication technology would be required." Some study of the state of materials technology, however, revealed that lack of knowledge was not really the critical factor. In many universities and private research laboratories, significant progress had been made in applying new theories of the behavior of materials to the problems of creating new and better fabrication processes. In addition, in certain materials fabrication shops, ingenuity, science, and some luck had combined to produce a very specialized capability in fabricating particular products out of particular metals. This knowledge about fabrication techniques was, however, diffuse; few if any firms had a broad capability in the area.

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[1]The term "special metals" refers to the refractor metals plus the super-strong alloys of steel (used primarily in aircraft and jet engines), titanium, beryllium, and dispersion modified materials (the specialty of the Du Pont Metals Center). The term "exotic materials" includes all of the above plus plastics, fiber-reinforced plastics and the newly developed composite structures that contain both metallic materials and non-metallic materials.

The Fabrication Barrier. "I just couldn't understand how so many things could be wrong with this industry," Mr. Hayes recalled. Seeking an answer, he began to engage in a personal program of research in which he visited metallurgy laboratories in universities, called on aerospace engineers, and talked to fabricators and other refractory metal manufacturers. What he found he called the "fabrication barrier." He explained:

> As a result of my survey, one fact stood out: all too frequently the intrinsic capabilities of these metals were being badly eroded by the time the end product was put to use.
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> At both ends of the distribution system from ore to product were highly science oriented firms. At one end were those like Fansteel, specialists in the extraction and consolidation of the metals. At the other end were large science oriented users like North American, Boeing, General Electric, Du Pont, and Westinghouse, where much of what they are doing, or plan to do, is a function of materials limitations.
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> But in the middle were the fabricators - casting specialists, machining experts, and forging professionals who were getting orders from the big prime contractors, buying the materials from us, and fabricating the product. These people were chiefly manufacturing oriented and were not in comfortable communication with the scientific community. Underutilization of the inherent advantages of the so-called exotic materials could be the only result.

Thus, Mr. Hayes found that despite the pressure from metal producers to expand utilization, and despite the need in many quarters for metals with the properties of refractories, the fragmented craft-oriented fabrication industry acted as an effective barrier to these pressures.

The Fabricators. The special metals fabrication industry worked on a "make-to-print" basis, with the design of each part or assembly being unique. As Mr. Hayes put it, "engines for jet aircraft do not have standard turbine wheels, nor do aircraft use standard wings. The fabricators, whether large or small, are necessarily job shops."

Although these shops had, for the most part, come into existence due to some special manufacturing capability, they were typically unable to develop or assimilate new technology. When, for example, several new firms appeared with an improved technique for making forged jet engine parts, the large forging companies were unable to adapt to the new technology and lost almost all of this business. Similarly, when a new technology for casting complex parts was developed by a number of firms, the major foundries, who were the logical beneficiaries of this innovation, lost this business to the new specialized firms. In short, Fansteel's management was convinced that the fabricators were typically too small to support significant development efforts, tended to

focus heavily on manufacturing activities rather than market opportunity, and were thus trapped into a pattern of temporary success followed by decay as their initial technological advantage dissipated.

Materials System Management. "When I first perceived the fabrication barrier," Mr. Hayes recalled, "I knew I had struck gold. Here was an opportunity ideally suited to Fansteel. We could integrate vertically into fabrication and horizontally across fabrication technologies and become known as the company that can make anything out of metal."

Fansteel's annual report to its stockholders for 1966 was in large measure devoted to describing the new strategy that had been developed. A key portion of Mr. Hayes' report is shown below:

> In Fansteel we believe that growth of the company will result if we do these things effectively:
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> 1. Participate in "materials sensitive" markets. The performance and quality of certain manufacturers' end products are highly interactive with the characteristics of materials which go into the product and processes used to bring materials to end product assembly. Thus, the manufacturer becomes materials sensitive. He earnestly seeks the accomplished materials supplier, and responds strongly to demonstrated capability and performance when he finds it.
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> 2. Develop a broad scope of related materials technologies. The objective is to provide, through horizontal and vertical integration of technologies, the interrelated capabilities which make it possible to convert "space age" materials into useful products and components within the walls and control of Fansteel.
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> 3. Build an organization rich in creative talent.
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> 4. Live with the key problems of the markets we serve. Sometimes companies can invent their way into growth. But the more certain path for Fansteel is to build growing sales volume in response to, and solutions of, specific problems to be found within present customer activities.
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> Combining these four guide-lines for growth produces the following fundamental philosophy:
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> Fill the role of "Material Systems Manager."
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> The complex problems of "materials sensitive" markets almost always require that a variety of material technologies be integrated in order to achieve an effective solution. The ability to perform this material systems management function is Fansteel's key to expanding opportunities.

The Plants of ASD

On a trip to San Diego in 1963, Mr. Hayes met Mr. Frederick Fletcher, who was at that time managing a metal parts fabricating company in the area. Impressed with Mr. Fletcher's engineering and business acumen, Mr. Hayes offered him the job of helping create a new division of Fansteel that would provide fabricated parts and structures to the aerospace industry. Mr. Fletcher was immediately attracted by Mr. Hayes' vision of bringing together a host of different materials fabrication technologies and agreed to take on the task.

Both Mr. Fletcher and Mr. Hayes believed that the most efficient way to enter new markets and technologies would be through acquisition, but in 1964 Fansteel's stock price was not sufficient to support an ambitious program. Consequently, they decided to proceed slowly and to make the initial acquisitions with cash.

History of Acquisition. In 1964, Fansteel acquired the BMW Manufacturing Co., a Torrance California based metal-working firm with $4 million in sales. BMW had a reputation as being one of the most skilled specialty metal fabrication job shops on the West Coast. Mr. Hayes quickly capitalized on this capability by moving a chemical equipment fabrication activity from the Metals Division in North Chicago to a new location close to the BMW plant. The combined facility was known as Fansteel/Torrance.

Airtek Dynamics Inc. was acquired early in 1965. This firm engaged in aerospace machining and specialized in the fabrication of titanium pressure vessels. At the time of its acquisition, Airtek was in financial difficulty due to a downturn in the aircraft metal parts shaping business, almost all of its competitors having already been forced out of business. Mr. Fletcher explained that by buying this company in a downturn, ASD was able to get their sophisticated pressure vessel technology at a reasonable price.

These two firms formed the core of ASD for the next two years. During this period, the Powder Metallurgy Plant was created in Harbor City, California, by moving engineers and machinery from the Metals Division. The purpose of this facility was to develop new processes for forming complex parts out of metal powders.

In 1967 the price of Fansteel common stock rose sharply to about $60 a share, representing a price-earnings ratio of about 30. Mr. Hayes believed that a significant portion of the trading activity had been due to institutional investors who had been impressed by the company's recent earnings growth and long term strategy. Suddenly having the currency to embark on a serious program of acquisition, top management moved quickly. By the end of the first nine months of the year, five acquisitions had been made and three others were in various stages of completion. Added to ASD were: Advance Casting Corp., Reflective Laminates Inc., San Fernando Laboratories Inc., and Precision Sheet Metal Inc. The agreement with Reflective Laminates also included an option to purchase Fiber-Tek Inc., a recent spin-off.

Also in 1967, ASD purchased the machinery and recruited the labor necessary to start a business in the machining of compressor blades for jet engines. This facility was known as the Jet Engine Plant.

Information on the managers, products and capabilities, type of technology, potential for expansion, sales, profitability, and growth of each of the business units of ASD is given in Exhibit 5. Exhibit 6 shows, again for each of the plants, the proportion of its sales which derive from the seven major markets that ASD served. The plant's fabrication capabilities, in terms of performing specific types of operations on various metals, are displayed in chart form in Exhibit 7.

Common Characteristics. The acquisition program had been undertaken to provide the division with new market positions and new technical capabilities. Despite the variety of firms that had been acquired, all of them had a great deal in common:

1. All of the firms were located in the Los Angeles area and served either the aerospace, chemical equipment, or nuclear energy equipment industries.

2. All of the firms could be classified as small or medium-small businesses, with annual sales running from $500,000 to $9,000,000.

3. All of the firms were engaged in selling a skill in materials fabrication rather than a line of standard products; i.e., they were job shops.

4. All of the firms were, at the time of their acquisition, managed by their original owners and founders. In November 1967, except at Airtek, these men were still actively managing their firms.

5. The pattern of operations in the plants was similar: a salesman would become aware of a particular piece of business and communicate this information to the plant manager. The engineers or production managers would then decide how such a fabrication would be best performed and an estimate of the cost would be made. A sample product might be produced and a bid would be submitted to the customer. If the firm's bid price was lower than that of its competition, or if there was reason to believe it could provide better schedule or product performance, it won the job, and a fixed price agreement was made for a certain number of parts or structures. The items then were fabricated, checked for quality, and shipped to the customer.

6. Usually, the design of the part or structure to be produced was specified by the customer. The engineers working within the individual plants, while well versed in the art of how to use different fabrication techniques to achieve a given end product, were typically not familiar enough with the technology involved in the end use of the product to develop the product design specifications.

7. All of the firms were subject to the cyclical nature of the aerospace business, since all were involved, to some extent, in serving this market. One of the reasons for the existence of the small fabricators was that the large prime contractors could not afford to support internal fabrication capabilities during the periods when their own business was slack. The independent supplier could, by contrast, sell his capability to whichever firm was currently busy. There were times, however, when the whole industry would experience a slowdown and the fabricators would have to cut back quickly on expenses, close temporarily, or go out of business.

8. With the exception of San Fernando Laboratories and the Powder Metallurgy Plant, none of the plants were engaged in science oriented development work. Their technical skills were often substantial, but were typically based more on "art" than on science. It was in craftsmanship,rather than science, that their critical skills lay.

The Materials System Management Function

Building upon Mr. Hayes' over-all philosophy for the development of the company, Mr. Fletcher defined the primary mission of ASD as "performing a materials system management function." Describing the meaning of this concept, Mr. Fletcher wrote:

> In order to achieve the improved performance and reliability which are increasingly demanded by the materials sensitive industries, and of which the advanced materials are capable, it is absolutely necessary to establish integrated technologies, closely relating what is done in one process step to its effect on all other process steps and the end product. To establish positive control of processes such as extraction, consolidation, and various fabrication technologies, it is necessary to perform a materials system management function.
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> To date, equipment builders, such as aircraft companies, try to perform this function by stringent specifications and procurement controls. Undoubtedly it is the best means available at present, because the strata of companies from which they purchase materials and components are clearly unable to perform the function themselves. In part, this is because the advanced materials industry is organized around functions. But it is also because the fabrication companies which supply these markets typically have a low order of technical capability, and are unable to comprehend and implement a materials system management approach.
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> The system of customer management of the manufacturing process sequence is inherently deficient. Simply stated, the separation of the management control function from the responsibility for manufacturing a product induces great problems of communication, conflict of interest, etc. With the increasing complexity of the materials systems to be used, and the ever more demanding environmental conditions, the problems and communication gaps are compounding.

> This, then, is our opportunity: to establish ourself as a material system manager, responsible for bringing critical components and materials to the market more efficiently, and with better quality and performance than would be possible within the existing industrial structure.

Business Activities and Policies

The policy of supporting, rather than controlling, the plants had been recently reiterated by Mr. Fletcher, primarily for the benefit of the managers of the new acquisitions. Support activities included help with hiring and personnel problems, a divisional sales force that brought extra business to the plants, the provision of the financial and planning resources necessary to undertake major business improvement projects, and aid in solving pressing temporary problems.

One plant manager complained that division management did not become involved in plant operations even when, in his opinion, it was necessary to pull together a deteriorating situation. "This type of response is to be expected," said Mr. Fletcher, "especially when a manager sees another plant gnawing away at the profits he worked so hard to produce. It is my conviction, however, that the long term interests of the division are better served by encouraging independence and self-reliance." Mr. Fletcher went on to point out that it was all too easy to give too much support:

> Early this year we had an opportunity to increase the level of sales and profits at both the Fansteel/Torrance and Airtek plants by taking on a large amount of metalworking involving long production runs of aircraft parts, tools, and structures. The new billings about doubled the level of operations at both plants and brought them close to their capacities.
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> It soon became apparent that serious losses were occurring. At Airtek there was a combination of poor estimating and higher than expected production costs. At Fansteel/Torrance, we learned that about the only thing that the people there can't do is to make a profit on long production runs - they are strictly experts in making small lots of unique items. As a consequence, we cutback the level of operations at both plants and imposed a 5-10% temporary ceiling on annual sales growth.

Management concluded from this experience that "more of the same" was often not a valid way to growth in job shops. Mr. Lee, Fansteel's vice president for management information and control, summarized the revised policy as "supporting consistent well defined projects, initiated by the plant managers, rather than supporting the plants themselves." He added,"There is usually a reason why a job shop is a certain size - it may have saturated a specialty market or, more commonly, it may have run into diseconomies of scale. In these cases, we don't want to expand the shop itself, but help the manager expand his horizons and do new things."

One example of the new policy was the vacuum casting project at Advance Casting. Both Mr. Wolf and Mr. McClellen, the managers and previous owners of Advance Casting, stated that they would probably not have tried to move into this new area of casting technology on their own. ASD had provided the funds and located and hired the engineering talent necessary to build and operate the new facility. Mr. McClellen enthusiastically endorsed the project, stating that "when the facility is complete, we will be able to do high quality casting in the superhard steels. Currently, there is only one other firm west of the Mississippi that has this capability."

Production Shops. "There is a distinct economic advantage to long production runs," Mr. Fletcher said. "In most of our shops we are turning out items by tens or twenties, or even by ones or twos in the case of space hardware. If we can get continuity of runs, like making 3,000 turbine blades, we will have highly expandable businesses. In addition, if the technology is closely held, the profitability can be much higher than the 10% made by the average shop."

Management had singled out the business of manufacturing parts and structures for jet engines as the most promising field in which to create production shops. Total sales in the U.S. of jet engines were expected to double in ten years, with three major engine builders accounting for 90% of the market. These firms tended to act as general contractors, contracting out the fabrication of parts and subassemblies, with the trend being towards subcontracting larger portions of the engines to fewer firms. Becoming a major fabricator for the engine builders was one of Mr. Fletcher's key goals for ASD: "It is the aircraft and jet engine boom that will really make Fansteel a growth company."

ASD management had an approach to this market based upon the development of a broad spectrum of capabilities pertaining to the production of jet engine parts. At Advance Casting, the new vacuum process would permit the casting of turbine blades and other engine fixtures out of superalloy steel. The Powder Metallurgy Plant engineers were in the final stages of developing a method for making titanium and superalloy parts out of metal powders that could be applied to jet engine part production. Precision Sheet Metal was engaged in making turbine blades out of sheet metal and the Jet Engine Plant used numerically controlled machine tools to make jet engine parts. The variety of engine types, and of environmental conditions within a single engine were so great that each of these techniques could have a role.

Management did express some concern over ASD's ability to be an efficient producer, seeing in the poor performance of the Jet Engine Plant and the troubles at Airtek and Fansteel/Torrance evidence of weakness in manufacturing management. "I am looking to a recent acquisition, Precision Sheet Metal," said Mr. Fletcher, "for leadership in this area. The people there have a demonstrated ability to run an efficient operation and I plan to hand over to them the facilities and responsibility for the performance of the Jet Engine Plant."

New Materials and Techniques. Part of the new corporate philosophy was the assertion that the fabrication barrier could be overcome without engaging in basic research - more important was the drawing together of state-of-the-art skills and knowledge. As the state-of-the-art improved, Fansteel could adapt through acquisition, always maintaining the advantage of an ability to apply and integrate knowledge. Within ASD, therefore, the emphasis would be upon extending current techniques, coordinating the capabilities of newly acquired firms with those currently in the division, and building profitable businesses out of scientific knowledge developed elsewhere.

The acquisition of San Fernando Laboratories (SFL) was an illustration of the policy of acquiring scientific knowledge and techniques just at the time they were ripe for practical application. This firm was basically a research lab that had developed the most advanced methods in existence for applying "chemical vapor deposition processes" to practical parts fabrication. This technology allowed a thin coating of almost any metal, including the refractory metals, to be applied to a heated form. It could be used to make such items as pure tungsten tubing more economically than could the Metals Division with its current methods. It also could be used to coat parts produced elsewhere in ASD with refractory metals. Mr. Holzl, the manager of SFL, explained that the lab had been sold to ASD "because we didn't see how we could handle the inevitable explosive growth of our production activity without the backing of a major company. I see the total revenues attributable to installing this process in various parts of Fansteel as growing to, perhaps, $10 million in five years."

Similarly, Fansteel's option to purchase Fiber-Tek in 1968 illustrated management's "show me" attitude towards new technologies. While Mr. Fletcher fully expected Fiber-Tek to successfully manufacture its special carbon fiber products and sell them to aerospace companies, the option provided protection against the major market and technical uncertainties.

Additional opportunities existed in combining some of the advanced technologies held individually by the plants. Mr. Fletcher gave this example of how new business could be created by integrating technologies:

> We have recently become aware, through the Metals Division, that the Air Force wants very badly a certain type of composite structure. On one hand, this structure is composed of carbonaceous fibers, and on the other it is infiltrated with a metallic matrix. With both San Fernando Labs and Fiber-Tek, and some development work, we can make this composite structure. So here is a case in which we can bring diverse technologies together to capture a piece of business that probably no other single firm in the world can get. I am now working with both Mr. Brown and Mr. Holzl on this opportunity.

Manufacturing Coordination. The development of an "integrated" manufacturing capability was necessary, according to management, if ASD was to make full use of the capabilities it had assembled. Mr. Hayes considered this process so important that to help him in the initial stages of building ASD he used a "shopping list," consisting of the various manufacturing techniques needed to make certain types of refractory metal aerospace structures. Management stated that in 1967, over 20% of ASD's sales involved products requiring the use of more than one fabrication technique.

Mr. Fletcher cited a recent contract for the production of special parts out of tungsten powder as an example of a coordinated manufacturing effort. The Metals Division supplied the raw tungsten oxide to the Powder Metallurgy Plant, where it was treated to obtain pure tungsten powder. There it was alloyed, pressed into shape and heat treated to give it strength and integrity. Fansteel/Torrance then finished the parts on machines of its own design. Neither plant, Mr. Fletcher stated, would have been able to do this work on their own. Thus, business had been created by bringing together relatively standard techniques - no technical innovation was required.

The potential for a similar type of cooperation existed between San Fernando Laboratories and Fansteel/Torrance. Chemical valve parts, machined by Fansteel/Torrance could be coated with tungsten or tantalum with CVD processes at San Fernando Labs. Management was undecided as to whether the parts should actually be shipped between plants, or whether the CVD process should be installed at Fansteel/Torrance.

Pointing out some of the problems associated with managing this type of activity, Mr. Hill, ASD's marketing manager, said:

> High margins and high profits do not always go hand in hand. We can often make very high margins on very special or unique items that require engineering or manufacturing inputs from several plants. This type of activity generates prestige and bypasses most of the competition, but without a high absolute level of profits, we can't payback our investment. I feel that ultimately our success will depend upon our ability to manufacture efficiently, in volume, particularly on projects that require inter-plant coordination. However, since most coordination is currently on an ad-hoc basis, I am not sure we can handle a higher volume.

Division Management

The main participants in the over-all management of the division were Mr. Frederick Fletcher, general manager of ASD and vice president of Fansteel, Mr. William Hill, marketing manager, Mr. Rolfe Wyer, division controller, and Mr. Jack Chelius, manager of design engineering. Mr. Robert Lee, who held the position of vice president of Fansteel in charge of management information and analysis, while not a formal member of the division's executive structure, spent close to one-half of his time at ASD, working closely with

Mr. Fletcher on matters involving planning, analysis, and policy. Since Mr. Hayes spent about one-third of his time on the West Coast, he maintained an office at ASD headquarters and was often available to members of the division's management for consultation on matters of policy as well as problems in operations.

Mr. Fletcher. The position of division manager immersed Mr. Fletcher in a myriad of activities ranging from the setting of long term strategy to hosting important visitors on tours of the plants. In the area of strategy, his chief concern was the orderly development of a capability within ASD for the performance of the "material systems management" function. On a level closer to operations, Mr. Fletcher devoted his attention to specific problem areas in the plants and acted as a focus for the exchange of technical information and the identification of business opportunities. Interaction among the plants was primarily between the technically oriented people, and Mr. Fletcher's reputation of being extremely knowledgeable in fabrication technology enabled him to win the respect necessary to function as a technical coordinator. Interaction on the business side of the plant's activities was somewhat weaker; Mr. Fletcher said that he could not recall any meeting that had been attended by all the plant managers. In order to keep in touch with the plant managers and help them in any way he could, he had adopted a policy of setting aside 25% of his time for meetings with them. Ordinarily, this meant that he spent a day or two each week driving from plant to plant in order to cover the whole division.

Mr. Hayes recognized that the time pressure on Mr. Fletcher was severe and stated that:

> Fred (Fletcher) needs a man to help him on the division level; he and I both are aware of this. What we see is someone whose job is almost to do nothing but identify specific business development opportunities at ASD. There are many opportunities available in synthesizing some of our activities, but you have to have someone worrying about how to do it.

Marketing - Mr. Hill. The marketing activity carried on at ASD would most appropriately be termed "two level selling." Each of the plants had a few full time sales personnel who reported to the plant manager and several field representatives. The plant salesmen were moderately well versed in the fabrication technology offered by the plant, with their chief competence being a knowledge of some of the purchasing agents and engineers within their market industries who had needs for the type of products their plant provided. The field sales representatives were not employees of the plant, but represented its interests in parts of the country that were not conveniently or often reached by the regular salesmen. When an important order was pending, the plant manager or chief engineer would usually become involved in talking with the customer and in negotiating a price.

On the division level, Mr. Hill operated a separate marketing group with six full time field salesmen. He described the function of his group as "increasing the profitability of the plants by providing a sales group that handles the whole range of skills and technologies available from ASD." To accomplish this he relied on two concepts: "customer oriented selling," and "skimming." By "customer oriented selling," Mr. Hill meant that his salesmen were assigned to particular customers or groups of customers, rather than to plants within ASD. There was, for example, a "Boeing man," who lived in Seattle, and by spending most of his time at this one firm, became very knowledgeable in the various types of work that were available there and established continuing relationships with the engineers and purchasing people. The division salesmen were typically not well versed in the details of the fabrication technologies offered by all the plants, but Mr. Hill believed that, despite this, they performed a valuable function:

> From a marketing point of view, these men can get the job done. They may not be able to go into a company and design plastics into their engines on the spot, or sell powder metals where they have never used them before, or any of the other things a technical specialist might do. But when it comes to getting an order at the best possible price, or getting an opportunity to take an order, he has connections that are more valuable than those of any specialist. He can always call in a specialist if he needs one.

By "skimming," Mr. Hill meant the process of making available to the plants so much business that the managers could afford to pick and choose among the jobs offered for those that would provide the greatest profit. This concept, he said, was particularly important in job shops that had definite capacity constraints, as their profits were strongly dependent upon the mix of size and profitability of the jobs they undertook.

Control - Mr. Wyer. Joining ASD in the summer of 1967, Mr. Wyer had been given the complete responsibility for developing and administrating a control system for the division. Mr. Wyer noted that the present control system was "very underdeveloped," and described his concept of the job ahead of him as "building an improved financial reporting system that will provide division management with information on the operations of the plants, and supplying a controllership function to the shops."

Each of the separate plants possessed its own internal reporting and control system. These varied in sophistication from a complex computerized job costing system at PSM to what was essentially a "checkbook" record keeping system at Advance Casting. With a view to the control and scheduling problems that would arise as ASD's volume of business grew, management was convinced that the division would have to provide aid to the plants in developing improved scheduling and financial reporting systems. Some of the steps Mr. Wyer planned to take in the coming year were the centralization of the following functions: (1) billing, (2) matching of payables and approval of payable invoices, (3) preparation of data on labor distribution, and (4) the administration of control reports.

Engineering - Mr. Chelius. The centralized design engineering activity, managed by Mr. Chelius, nominally provided engineering services to the entire division, but actually focused on Fansteel/Torrance, Airtek, the Jet Engine Plant, and, to some extent, the Powder Metallurgy Plant. The group had originally been composed of chemical equipment design engineers who had been transferred from the Metals Division when that business area was moved to Fansteel/Torrance. Drawing additional talent from Fansteel/Torrance, the group expanded its talents into generalized tool design and stress analysis. Coordination with engineers at Airtek and the Powder Metallurgy Plant had added even greater diversity.

Mr. Chelius indicated that although his group was beginning to aid some of the newly acquired plants, it did not really possess the broad range of capability required to serve them. "We can do tool design across the board," he said, "but in some of the fields the new plants are involved in it is difficult with the staff I have to come up with anything productive. I don't really know where I could get the talent I would need to work in these areas."

Organizational Issues and Alternatives

"There exists at ASD," commented Mr. Fletcher, "a long term problem of structure. Viewed in the abstract, I have to find ways of both preserving the autonomy of the plant managers and promoting close coordination among disparate parts of the division. In addition, there are different types of coordinative activity possible, emphasizing either the technical, manufacturing, or marketing areas, and I have to find the right balance among them. Ultimately, I think that these types of conflict, between goals of autonomy and integration and between the need for both efficiency and mastery of complex technologies, are never really resolved - they are managed.

"In the near term, one of the first steps I want to take is to make ASD easier to manage. I believe this can be done by clustering the plants into groups and creating a new level of management between the plant managers and myself."

Grouping the Plants. It was the pressure of growth that prompted Mr. Fletcher's suggestion of grouping the plants. He envisioned ASD as growing both through the expansion of the businesses in which it was already engaged and through the acquisition or creation of new businesses. While division management was fairly certain that several of the business activities at ASD had the potential for rapid growth, there was concern over the expandability of the individual plants. Mr. Lee observed that he had "never seen a really successful aerospace fabrication operation that was larger than could be readily managed by one man on a day to day basis." Agreeing with this principle, Mr. Wolf (manager of Advance Casting) said:

Our success in my business depends upon giving the best possible service to the customer. There have been times when I would put a finished order in my car to deliver it personally on a Saturday night. If we were to, say, triple the size of our operation, this type of service would not be possible. The smaller shops would get the most profitable business and we would face the possibility of really getting killed in a downturn.

Recalling his experience with trying to double the sales of Fansteel/Torrance and Airtek, Mr. Fletcher said: "I don't plan on taking $4 million job shops and turning them into $8 million job shops, I have already been up and back down that road. What I think we can do is grow by units and grow by expanding production shops, rather than job shops."

Since growth meant the addition of new plants, Mr. Fletcher believed that grouping the plants was the most logical method of accommodation. How these groupings would be constituted was a major organizational decision. The scheme he favored was to split the division into three parts, with the differentiation being based upon the nature of the fabrication technology involved. At the time of the case, he had tentatively decided that these groups should be "metalworking," "direct forming," and "nonmetallics," and should be initially formed as follows:

Metalworking	Direct Forming	Nonmetallics
Fansteel/Torrance	Advance Casting	Reflective Laminates
Airtek	Powder Met. Plant	(Fiber-Tek)
Precision Sheet Metal	San Fernando Labs	

The Jet Engine Plant is not explicitly shown because Mr. Fletcher planned to combine it with Precision Sheet Metal. The inclusion of Fiber-Tek in the Nonmetallics group was, of course, contingent upon its acquisition in 1968.

The group managers placed in charge of these clusters of plants would have the responsibility for the over-all development of the businesses involved and perform such tasks as capital budgeting, planning and execution projects requiring the coordination of the the plants in their group, helping to coordinate activities between groups, and managing the addition of new plants to their groups. The plant managers would retain complete responsibility for the profitable operation of their separate businesses.

This particular rationale for partition was based upon the fact that the "metalworking" plants were chiefly engaged in changing the shape of a piece of metal by bending, cutting, welding, machining or otherwise applying forces to an already consolidated material. The "direct forming" plants, on the other hand, attained the desired final shape in a single step by directly consolidating molten, powdered, or vaporized metal into that shape. The "nonmetallics" group did not deal with metals at all. Mr. Fletcher expected that this grouping plan would make it easiest for the group managers to build teams of skilled engineers whose capabilities spanned the whole range of technologies within the groups.

One of the arguments expressed in opposition to this structure was that it was more important to cross technological boundaries between plants and between the division and the customer than to reinforce them. While this line of reasoning did not suggest any unique grouping, several alternatives were possible. One manager, for example, believed that the groups should be formed so as to separate the technically static from the technically changing plants, as follows:

Technically static	Technically changing
Fansteel/Torrance	Powder Met. Plant
Airtek	San Fernando Labs
Precision Sheet Metal	Reflective Laminates
Advance Casting	(Fiber-Tek)

This grouping tended to place the more science oriented plants together and to insulate the plants in which manufacturing efficiency was the prime consideration from those in which adaptation to changing technology was more important.

Management had also given some consideration to a grouping based upon common customers or markets, but had decided that such a structure was not attainable without destroying the integrity of many of the plants. As can be seen in Exhibit 6, there was a high degree of overlapping in the market interests of the plants.

Mr. Fletcher indicated that how he would eventually group the plants would depend not only on the type of arguments just discussed, but on the availability of suitable individuals to fill the positions on the group level. He had tentatively identified Mr. Brown (Reflective Laminates), Mr. Holzl (San Fernando Labs), Mr. Grien (Precision Sheet Metal), and Mr. Preston (Powder Metallurgy Plant) as those managers, currently in ASD, who had the ability and experience to be group managers. He noted, however, that it would be difficult for the plants involved to do without the leadership of these men on short notice.

The Product Manager Option. Mr. Hill, ASD's marketing manager, stated that although the plants could not be practically grouped according to common markets, a greater degree of market orientation was desirable. Accordingly, in October 1967, he reorganized the divisional marketing group, creating four product managers responsible for the business areas of powder metal parts, jet engine components, aerospace structures, and chemical equipment. Salesmen who had previously reported to the plant managers were directed to work through the appropriate product manager on each sale. It was the responsibility of the four product managers to coordinate between the salesmen, the design engineers, and the plant managers. Mr. Hill believed that this plan would benefit the plants in two ways: (1) It would result in a better allocation of the overall sales effort since it provided an organizational mechanism for freeing up sales people from plants where they were temporarily not needed, and putting them to work selling in areas that could immediately profit from the extra effort. (2) It would increase the salesmen's knowledge of all the fabricating technologies within ASD and lay the groundwork for an ability to market complex material systems.

By November, the reaction of the majority of the plant managers to this move was clear: it was unacceptable. The major objection was that it would hurt the plant's business to have salesmen representing it who were not familiar with the technology. In addition, several plant managers stated that product managers were not necessary and in fact impeded their ability to respond quickly to opportunities for bidding.

While Mr. Fletcher recognized the theoretical merits of Mr. Hill's plan, he expressed deep concern over the resulting conflict. He stated that "for the present I think that the best move would be to drop the plan and have the salesmen report back to the plant managers." Mr. Hill agreed that the current tensions would nullify any positive results that might be gained from his reorganization, but pointed out that "somewhere along the line, we are going to have to do more than just talk about integration."

The Plant Managers' Function. Both Mr. Fletcher and Mr. Hayes felt strongly that ASD should allow the plant managers complete control over all the important aspects of their businesses. Speaking to this point, Mr. Hayes said:

> When small business is successful it is usually due to the strength and entrepreneurial drive of a single key manager. Such managers have to be smart, aggressive and self-reliant just in order to survive. Man for man they will run right over the kind of manager you find in the typical big corporation. The challenge we face here in ASD is to build an organization based upon the talents of this type of person.
>
> The key to handling these firms properly is not to change the nature of their businesses. You are a fool if you try to change the nature of assets you have acquired. It is very important that the plant managers understand that they have the whole responsibility for performance. On the division level, the attitude must be "what can we give them?" rather than trying to direct them.

Most of the plant managers also had strong feelings on this subject. Mr. Woolson (Fansteel/Torrance) commented:

> None of the plants in ASD have products as such. What they do have are skills; I like to say that we live by our wits. Success is chiefly a function of working very hard to stay abreast of the local customer's needs, even anticipating them. If your wits are filtered by a big organization, you just cannot be effective in the market.

Despite all desires to the contrary, management realized that complete autonomy for the plant managers was not possible. The simple fact that their acquisition by ASD meant that their earnings would be pooled and rationed implied that there would be competition for resources. In making out the capital budget for 1968, for example, Mr. Fletcher had used a return-on-investment criterion for deciding to heavily support the development of vacuum casting at Advance Casting and to deny the request for replacement of ageing equipment at

Fansteel/Torrance. While this type of allocation decision was, management believed, essential to the growth of the division as a whole, it did have the effect of magnifying the differences among the plants. Businesses that previously served different markets with differing growth rates and levels of technology suddenly found themselves in competition with one another for the same pool of internal resources.

Mr. Fletcher was aware of the fact that competition for the division's resources might tend to frustrate the entrepreneurial drive of some of the plant managers, but thought that the problem was not serious. He explained:

> The more entrepreneurial managers, the ones who are more apt to chafe in an organization when everything is not under their control, are the same men we are counting on to develop new business opportunities. One of the main inducements we can offer such men to stay with ASD is certainly not the security of a big company, but the opportunity to work from a bigger base - to be able to do things they could only dream of doing before. We will not force, nor even allow, a good entrepreneurial manager to sit on a business that isn't developing. We have better uses for him.

This commitment to the value of the entrepreneurial plant managers, combined with an advocacy of their "hard-nosed" attention to detail and energy in pursuing their business interests was a recurrent theme throughout the management of ASD. However, in the opinion of one observer, this philosophy was not consistent with the company's philosophy:

> It is these very plant managers who are members in good standing of the fragmented fabrication industry, are production oriented in the extreme, and who run shops in which, for the most part, they are unable and unwilling to support any really innovative development efforts. ASD can't have it both ways - it can't infuse the shops with new technology or link them together horizontally without interfering somehow with the plant manager's prerogatives.

The whole issue of the role of the plant managers was further complicated by the fact that Advance Casting, and Reflective Laminates had been acquired on incentive agreements. Since the owner-managers would receive, over the next two years, additional amounts of Fansteel stock as a function of their profit performance, it would be unethical, Mr. Fletcher believed, to jeopardize in any way their ability to maximize their incentive gains.

Promoting Coordination. Although the members of ASD's management were in basic agreement on the goal of performing a materials system management function, there were significant differences of opinion as to how the coordinative activities should be structured. In speaking of the problem of relating activities in various parts of the division in order to make the whole more effective than just the sum of its parts, most people in ASD used the word "integration" to refer to this process. An independent consultant, who had worked with Mr. Hayes and Mr. Fletcher in identifying some of the organizational alternatives for ASD, noted that the several meanings attributed to the term "integration" was a source of imprecision and confusion in discussions among management. He explained:

One type of integration that seems to be desired is the combination of several different material technologies to produce a totally new capability within the company. If a need for this capability exists in the market, then a business development program would have to be initiated to manage a coordinated sequence of actions to bring the new technical knowledge, engineering talent, production process know-how, and marketing skills to bear on the problem of generating a profitable volume of new business. A case in point is the combination of the CVD technology of San Fernando Labs with the capability of Fiber-Tek to make carbonaceous fibers in order to produce a totally new type of composite material.[1] This type of integration is technical, innovative, and developmental in nature and emphasizes the coordination of engineering and marketing skills.

A second type of integrative activity is the coordination of state-of-the-art manufacturing capabilities at two or more plants in order to make a particular structure or part. An example is the parts produced through coordination between the Powder Metallurgy Plant and Fansteel/Torrance.[2] In this case the integrative leverage was not obtained from innovation, but from providing a lower cost of coordination than could be obtained from other materials suppliers or by the aerospace prime contractors. Here the tasks are perceiving and bidding on the job, coordinating the design of the production process, setting up schedules, allocating the responsibility for quality control, shipping the product, and deciding how to divide the profit. The essential nature of this type of integration is managerial and organizational, with the coordination between manufacturing operations being the primary problem. To obtain this type of business, ASD will have to find a way to achieve a lower cost of integration than its competitors.

The more technically oriented members of ASD wanted the integrative emphasis to be placed upon engineering activities. Most of the division's engineers, it was argued, were experts in production processes rather than product design. Since the full value of combining various fabrication technologies in one organization would only be realized if better products could be made, they felt it was necessary to increase the division's ability to engage in product design. As one plant manager stated the issue:

> The main problem in achieving a greater ability to bring the technologies of ASD to bear on industrial customers is to develop engineers who can deal with a variety of fabrication techniques in approaching the question of how best to make a product. The group structure that Fred (Fletcher) has proposed is necessary in order to develop this capability. The individual plants are too small to support a critical mass of people and at the division level the range of technologies is too broad.

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[1] See page 14.

[2] See page 15.

At the group level the technologies have enough in common and the ability to support an engineering team is sufficient so that a good capability to do product engineering can be created. These same teams would also have the skills necessary to help the plant managers within the group to install new production processes, improve their techniques, and expand into new business areas with complementary technologies.

Other managers agreed with the idea that the drawing together of engineering talent was a primary prerequisite for success, but, placing a greater emphasis on a customer oriented approach, believed a joint marketing-engineering team on the division level was needed. The fabrication barrier was due, they argued, partly to the customer's lack of knowledge of materials and fabrication capabilities. Consequently, the existence of a capability in ASD would not guarantee its use. Mr. Holzl, the manager of San Fernando Laboratories, explained:

Suppose a design engineer is unaware of our ability to coat metals with tungsten. He assumes that stainless steel will have to do and designs the whole system based on that assumption. When a fabrication contract is let, it is for a stainless steel part. We never even see the business and he does not get the product performance he could.

I used to work for a firm that had the same problem that ASD does. What we did there was to recruit technically strong people from the plants and make a sales group out of them. Often this group would invent on the spot for the customer. This was a very effective approach. For example, one time our sales manager pointed out to the customer's chief engineer how his engineers had made a subtle design error. The project engineer watched this performance, turned to me, and said "If this guy is a salesman, what are your engineers like?"

That, to me, is the essence of making a concept like Warren's (Hayes) and Fred's (Fletcher) work. I don't think we should wait around for new business to come knocking on the door. What has to be done is to centralize enough technical knowledge that we can go out and create business by solving our customer's materials problems for him.

Mr. Fletcher agreed that this concept had merit, but said that ASD might not presently have enough depth in engineering to implement it. "Most of our engineers are specialists," he pointed out, " and the 'tiger team' approach to marketing requires high competence not only in all the fabrication technologies at ASD, but in product specialties like electronics, jet engine design, rocket engines, airframe design, etc. As in many of our opportunities, the limiting factor here is people. Where do we get a guy who can do this job?"

Other members of ASD's management stressed the need to coordinate the abilities that already existed within the division, rather than increasing its ability to perform innovative and sophisticated technical work. Mr. Hill, the division's marketing manager, spoke of performing the material system management function this way:

> First, you have to distinguish between our normal single-plant type of business and business that involves inter-plant coordination.
>
> As far as the individual plants go, two thrusts are needed to make ASD work - you have to improve technology in some areas to keep sales and margins high. In those areas where you are not "discovering the atomic bomb" every week, the key is to keep margins high by being plugged into a large centralized sales force that can provide selectivity.
>
> When more than one plant is involved in a job, I believe that you need a "phoney" customer who stays between the real customer and the plants. You just can't integrate with "good feelings" - you have to have someone who will negotiate with the customer, negotiate with the plants, see that the layout is done, expedite the delivery of a sample, get the engineers together, and hold everyone to a schedule.

Mr. Hill went on to state that the division level was the most appropriate place for a surrogate customer to be created within ASD. There, he said, it could serve all the plants and maintain a one-to-one contact with the customer. Other managers, however, stated that the group managers, if and when such positions were in existence, should have this responsibility, as they could also provide a single internal point of contact for the customer and, being closer to the technical aspects of the task, could provide better engineering support. In the case of jobs that crossed group boundaries, it was argued, the division manager would have to decide which group manager would have the project responsibility.

Not everyone at ASD believed that the group structure was necessary. Mr. Lee, for example, stated that he favored an absolute minimum of centralization of manufacturing and engineering responsibility. He suggested that eventual centralization of the sales function might be desirable and that an inter-plant project group with profit responsibility could serve to coordinate other activities:

> My views on this matter are probably a bit more extreme than those held by other people in ASD. In time, I would like to see all the salesmen on a divisional sales force. The flexibility this would allow in terms of shifting sales personnel to meet peaks and valleys in demand in different areas would be quite valuable to the division. This would have to be done gradually so that everyone would understand how the system works and gain confidence in it.

The problem with any integrative structure is that a balance has to be maintained between centralized activities and the need for the plant managers to be able to act independently. I think this can be overcome by making the integrative unit a line, rather than staff, group and giving it P & L responsibility. Create a new business unit expressly charged with the responsibility of performing the material systems management process. The key managerial task of this group would be to reduce to a bare minimum the cost of coordination. This is the only way we will be able to compete in this area.

The integrative unit would deal with the market directly and seek out those orders which would involve the integration of more than one material technology. They would then work with the plants within ASD, or without for that matter, and subcontract, persuade, or whatever, in order to get the plants to participate. Such a group would have to have marketing, engineering, and management competence. Hold them responsible for earning a profit, let them compete for the output of the plants, and you will be able to see in black and white whether the concept is paying off.

Mr. Fletcher's Opinion

Responding to the question of how he thought ASD should be structured so as to function as a materials system manager, Mr. Fletcher indicated that he preferred to focus his attention on "activities" rather than formal "structures" and went on to state that he favored an entrepreneurial approach to the problem of integration:

Currently, ASD would best be described as a loose association of firms with an overlay of nontechnical selling skills on the division level. It is obvious that in order to function as a "materials system manager" we are going to have to build our competence to relate different parts of the division to one another. To look at the development of this competence solely in terms of "organization" is, however, naive. We are not operating a coal mine or running a steel mill - our central problem is not to tightly control people by dividing up tasks into little pieces, but to release people's energies and to learn to work with one another on tasks whose structures are not, a priori, very predictable.

Now, if you think about the "material systems management function" in terms of the kinds of things people will have to learn to do and the kinds of relationships that will have to develop, it is clear that the way in which the division will grow is a strong function of the type of activities in which we engage. Thus, I tend to view the question of organization in terms of the type of activities that are emphasized and how this emphasis changes over time. In essence, if I start by pushing in those areas that require coordination on the scientific and engineering levels, rather than stress efficiency in manufacturing and its coordination, the organization that develops will be different as to its style, sense of purpose, and ability to adapt to future change.

While efficiency is always important in a manufacturing business, I believe that this is primarily the responsibility of the plant managers. Division management can provide help, capital, and services to strengthen a plant, but it is not our function to operate the business. What we should, and must, do is to concentrate on the entrepreneurial aspects of business improvement.

I would like to have a small elite guard to push the integration concept within ASD. This may involve two or three people who are strong technical innovators; they have to be the very best we have. Their job would be to bring ideas and opportunities to me so that I can use the facilities that are assigned to this division to the best profit of the company. When such a business development project is completed, the actual operating responsibility would be assigned to one of the plants or a new plant might be created.

As far as the administrative portions of the business go - we have people on the division level whose task it is to standardize procedures so that ASD will be easier to evaluate and manage. They are not, however, to interfere in any way with the plant managers' ability to make a profit. The same goes for my elite guard; it is not their function to tell the plant managers how to run their businesses.

That is about as far as I see it going. I am not in favor of any huge complex of people as you will often see in the aerospace industry. I am talking about two or three people and an annual expenditure of maybe $100,000 on $50 million in sales. This will be an extremely small but effective commitment - the actual central spark of this concept of integration.

Exhibit 1

FANSTEEL: THE ADVANCED STRUCTURES DIVISION

Summary of Corporate Operating Results for the Years 1960-67

Year	Net Sales in Millions	Gross Margin on Sales	Net Margin on Sales	Ret. on Assets	Ret. on Equity
1960	$30.8	30.0%	6.18%	5.90%	8.5%
1961	32.3	28.0	4.14	3.56	5.52
1962	37.6	25.4	3.22	3.51	4.98
1963	33.8	20.2	0.21	0.22	0.30
1964	35.1	21.5	1.85	2.09	2.77
1965	44.2	22.4	3.43	4.49	6.02
1966	53.7	23.8	4.61	5.21	8.80
1967*	80.0	24.0	4.13	5.16	9.43

*Figures for 1967 are estimates based upon results for first ten months.

Year	Earnings per Share as Reported in Year Acheived	Earnings per Share Computed on Basis of 1967 Shares Out
1960	$1.39	$1.15
1961	0.96	0.81
1962	0.88	0.73
1963	0.05	0.04
1964	0.26	0.21
1965	1.12	0.92
1966	1.80	1.50
1967*	2.00	2.00

Source: Annual Reports and Company Officials

Exhibit 2

FANSTEEL: THE ADVANCED STRUCTURES DIVISION

Corporate Organization Chart

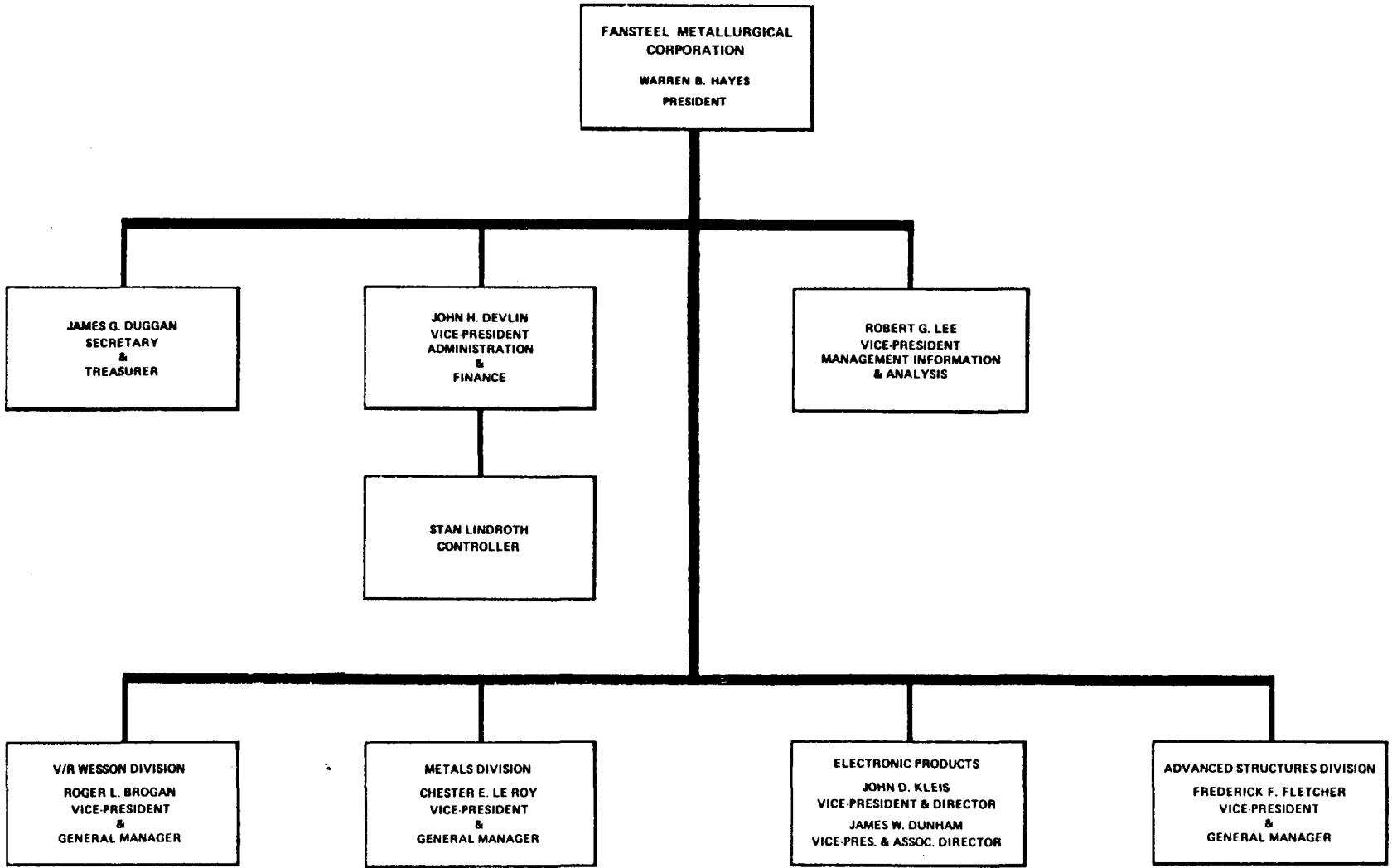
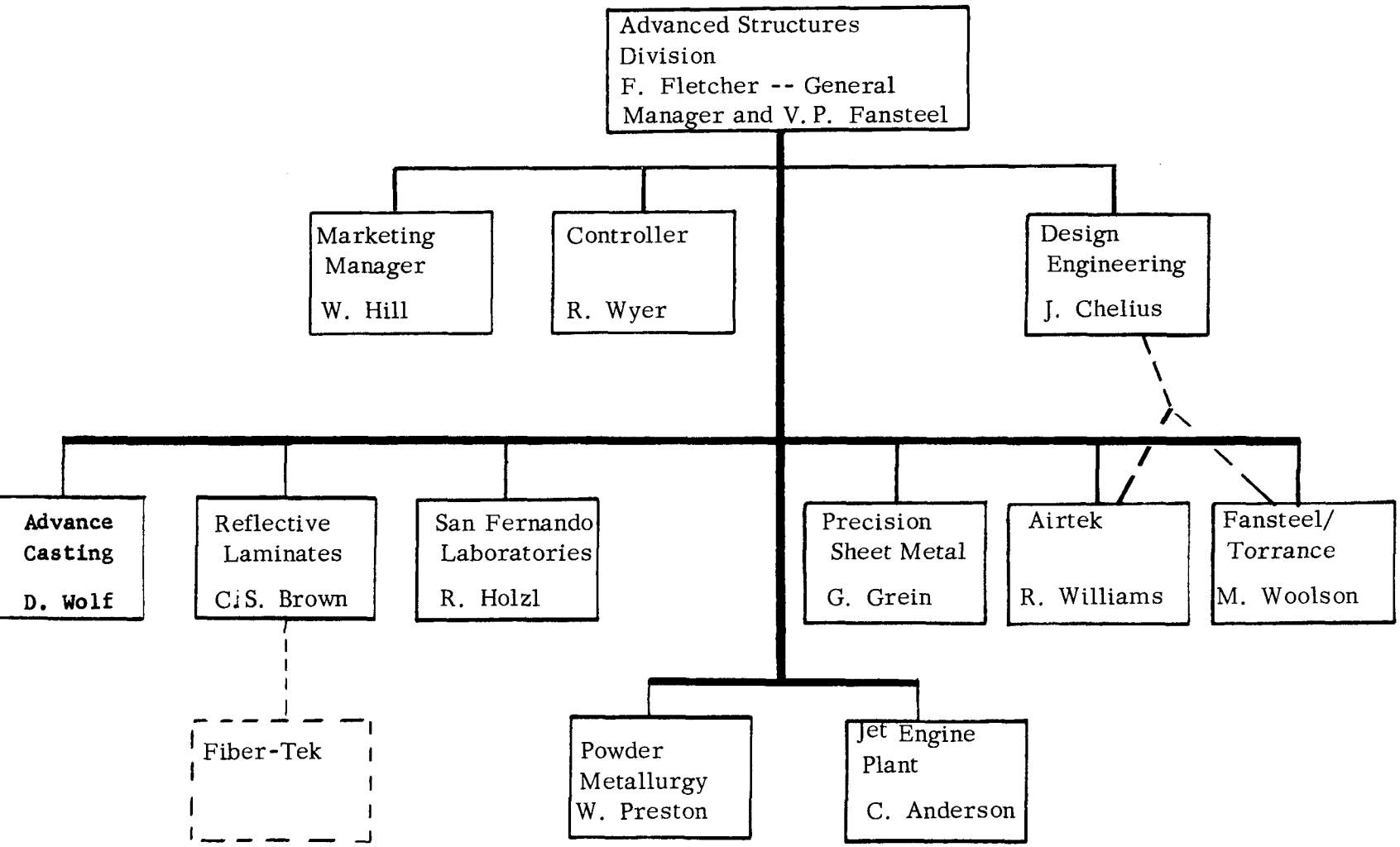


Exhibit 3

FANSTEEL INC. : THE ADVANCED STRUCTURES DIVISION

Organization Chart for ASD

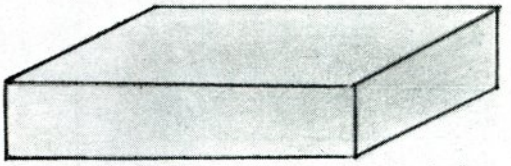


Source: Company Officials

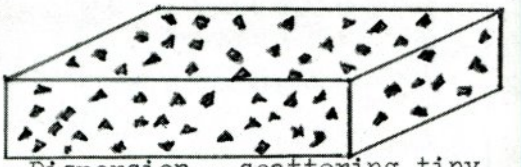
Exhibit 4

FANSTEEL INC. : THE ADVANCED STRUCTURES DIVISION

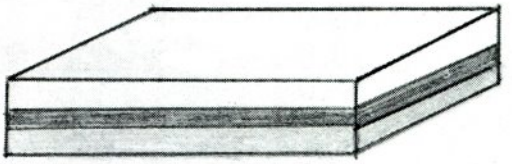
Ten Types of "Materials Systems"



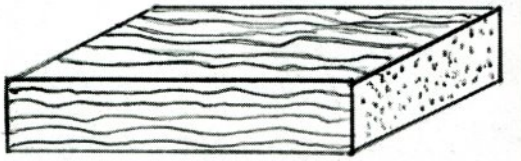
Alloying -- mixing one metal with another in the melt stage.



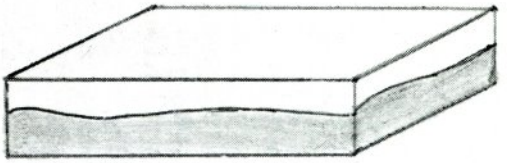
Dispersion --scattering tiny particles through the matrix.



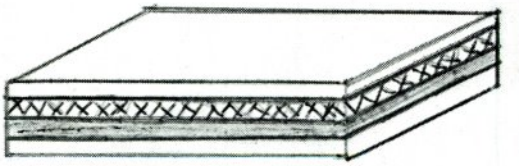
Bonding -- joining two materials by using a third.



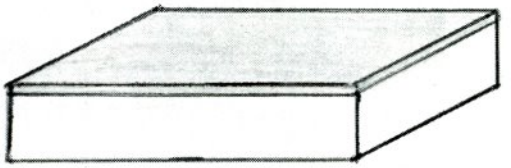
Fiber reinforcing -- embedding fibers in the matrix.



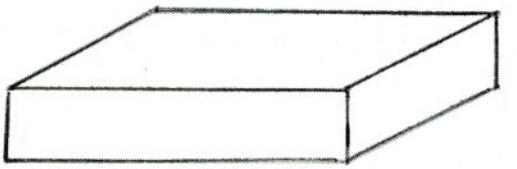
Cladding -- joining two materials to create a bond at the interface.



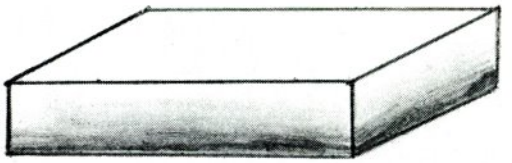
Laminating -- joining three or more materials together.



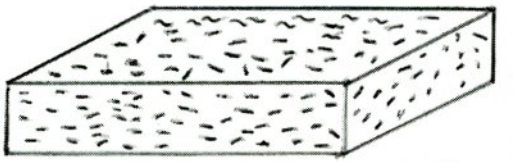
Coating -- applying a surface layer.



Powder compacting -- pressing powder to create the material.



Diffusion -- producing a more



Whisker composite -- single crystals

Characteristics of the Business Units of ASD

Business Unit and General Manager	Major Products and Skills	Character of the Technology	Potential for Expansion	Projected Sales for 1967	Pretax Margin for 1967	Recent Growth Rate in Sales
Fansteel/Torrance Mr. Woolson Acquired in 1964	Metalworking shop specializing in the production of small numbers of unique items. Skills include general machining, tooling, and precision fabrication in all metals.	This business uses large amounts of expensive capital equipment and highly skilled labor. The firm possesses unusually high skills in techniques that are relatively widely held.	Low, due to labor skills required and complexity of the job shop operations. Recent expansion resulted in losses before cutback.	$5,800,000	5%	Little growth prior to recent expansion, now limited to 5-10% per year.
Airtek Mr. Williams Acquired in 1965	Profiling (shaping) of aircraft parts from superalloy steels. Fabrication of ultra-light high precision titanium pressure vessels.	Widely held capability. Highly proprietary -- requires specialized equipment and skilled craftsmen.	Recent expansion of profiling activity resulted in losses. Pressure vessel business limited by demand.	$3,000,000	0%	Erratic
Jet Engine Plant Mr. Anderson	Machining of jet engine blades out of superalloy steels.	Numerically controlled machine tools. Many firms possess this capability, but peak demand still exceeds total supply.	Depends upon ability to operate profitability. Long production runs favor expansion.	$400,000	Loss	New facility
Precision Sheet Metal Mr. Grien Acquired in 1967	Fabrication to high precision of parts and structures from sheet metal, including the refractory metals. The firm specialized in making rocket nozzle extension skirts.	No other firm in the world has comparable skills in working with refractory metals in sheet form. The extension skirt fabrication equipment is unique.	Most of the business in current product areas is limited by demand.	$8,400,000	5%	Slight decrease
Powder Metallurgy Plant Mr. Preston Formed in 1966	Production of weights and vibration damping elements out of metal powder. In development is a new process for making titanium and superalloy powders.	Widely held -- highly competitive producing a relatively low profit. If successful, new process could provide considerable cost savings over alternate means of making parts out of titanium and the superalloys.	Highly expandable is new process development is successful	$2,600,000	Loss	Erratic
Advance Casting Mr. Wolf and Mr. McClellen Acquired in 1967	Air melt investment casting (lost wax process) of stainless steels. Major products are aircraft parts and valve bodies.	Investment casting allows the direct forming of complex shapes. While many firms possess this capability, few achieve the level of quality available from Advance. Scrap rate is much lower than normal	Expansion into vacuum melt casting is under way. There is some concern over maintaining level of quality with growth.	$2,300,000	33%	40%
San Fernando Lab. Mr. Holzl Acquired in 1967	Chiefly a development lab for chemical vapor deposition processes (CVD). Supports this work by producing parts for chemical and nuclear industries (e.g., tantalum tubing)	No other firm in the world has developed CVD technology to the level of control and quality attained by this firm. The development work is science oriented. Some patent protection exists.	Production portion of business could be readily expanded. In addition, many applications for CVD processes exist elsewhere in Fansteel.	$600,000	10%	20%
Reflective Laminates Mr. Brown Acquired in 1967	Fabrication of fiber reinforced plastic structures for aerospace applications (e.g., missile nose cones). Also produces thermal and acoustic insulation for aircraft.	While the principles behind this type of fabrication technology are widely known, few other firms are as skilled in their efficient application as Reflective. Not capital intensive.	Present growth is rapid and, according to management, is primarily limited by the firm's ability to accommodate.	$3,700,000	18%	60-70%
Fiber-Tek Option to purchase in 1968	A spin-off of Reflective Laminates, Fiber-Tek is developing a process to manufacture special carbon fiber yarn for aerospace applications	Little or no competition in this process technology exists. Other firms could develop comparable processes without great effort if market warrants.	Depends upon demand and cost level acheived.	---	---	---

Source: Company records and interviews with company executives.

Exhibit 6

FANSTEEL, INC. : THE ADVANCED STRUCTURES DIVISION

Proportions of the Plant's Sales Coming from Each of Seven Major Market Areas During 1967

Major Market Areas

Plant	Aircraft	Jet Engines	Missiles and Space	Military Hardware	Chemical Equipment	Nuclear Power	Other Commercial
Fansteel/Torrance	0.5		0.22	0.01	0.13		0.14
Jet Engine Plant		1.00					
Airtek	0.65		0.35				
Precision Sheet Metal		0.28	0.69			0.03	
Advance Casting	0.75		0.10	0.15			
Powder Metallurgy Plant	0.20		0.30				0.50
San Fernando Labs					0.60	0.40	
Reflective Laminates	0.40		0.44	0.04			0.12

Explaination: Read "0.5" at top left of table as meaning that 50% of Fansteel/Torrance's sales in 1967 were of parts and structures to builders of aircraft, for use in aircraft.

Source: Company records

Exhibit 7

FANSTEEL, INC.: THE ADVANCED STRUCTURES DIVISION

Capabilities of Plants in Terms of Applying Fabrication Techniques to Metals

Fabrication Technique	Metal: Stainless Steels	Superalloy Steels	Titanium	Tantalum	Columbium	Tungsten	Molybdenum
Machining	Airtek, Jet Engine Plant		Fansteel/Torrance, Prec. Sheet Metal				
Welding	Fansteel/Torrance, Airtek, Prec. Sheet Metal						
Forging							
Casting	Advance Casting	(in development at AC)					
Powder Metallurgy	Powder Metallurgy Plant		in development				
Sheet Metal Fab.	Precision Sheet Metal						little done
Rolling, extruding, etc., mill forming			little done	Metals Division of Fansteel			
Chemical Vapor Deposition			little done	San Fernando Laboratories			

Source: Interviews with company executives

Exhibit 8

FANSTEEL: THE ADVANCED STRUCTURES DIVISION

Photograph of Machining Operation at the Fansteel/Torrance Plant

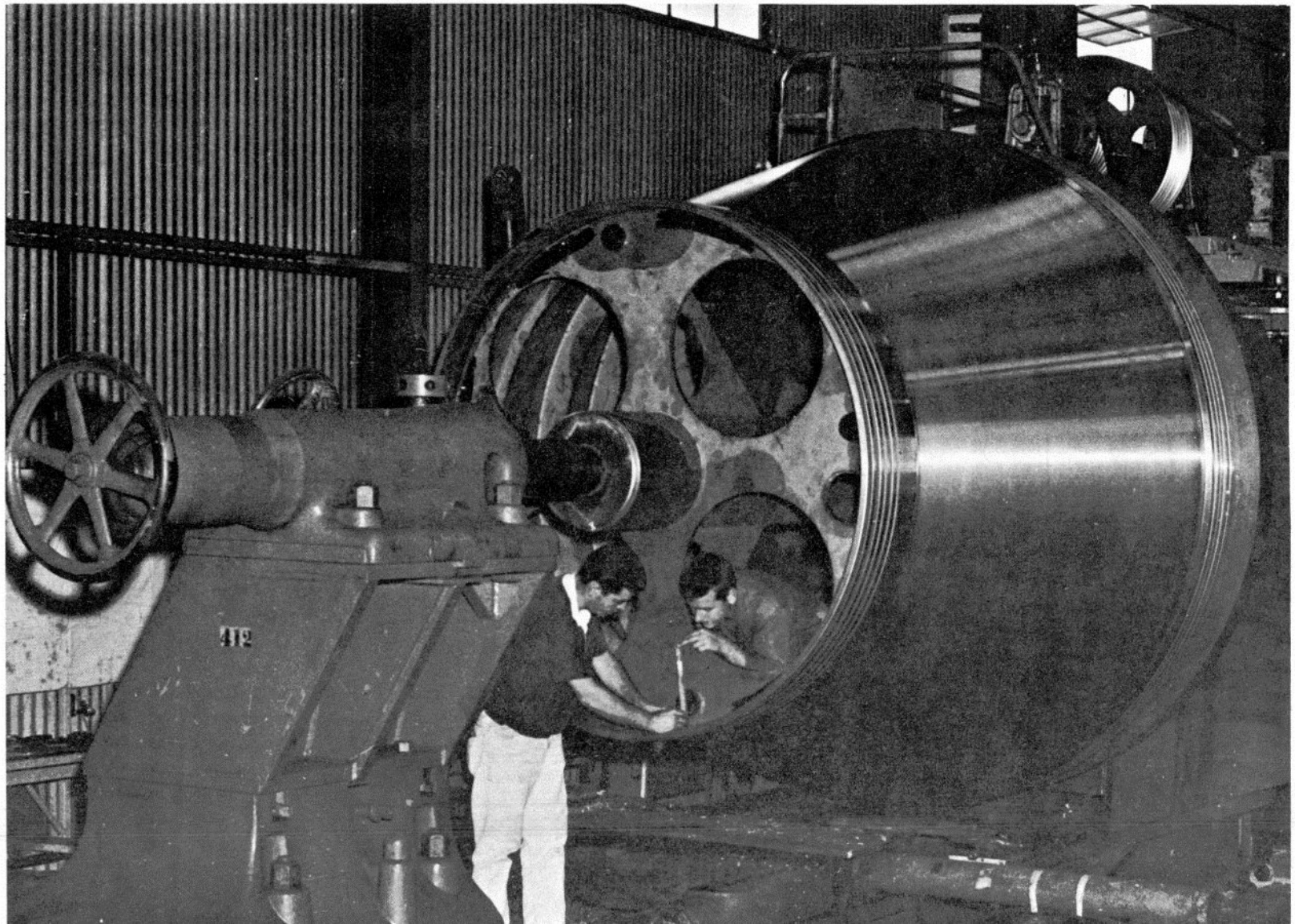


Exhibit 9

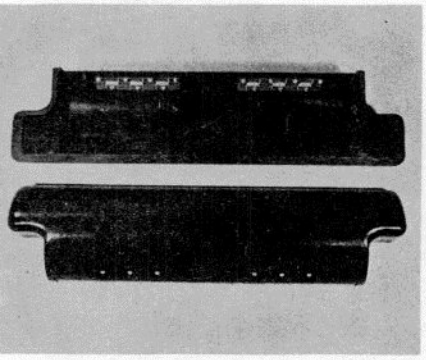
FANSTEEL: THE ADVANCED STRUCTURES DIVISION

Promotional Material Used by the Reflective Laminates Plant

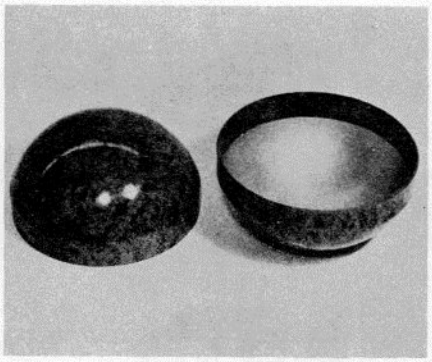
FABRICATION OF REINFORCED PLASTICS AND ABLATIVE MATERIAL INTO USEFUL STRUCTURES FOR THE AEROSPACE INDUSTRY IS A CAPABILITY IN WHICH VERY FEW COMPANIES CAN OFFER COMPLETE ONE SOURCE RESPONSIBILITY. THE CONTINUAL EXPANSION OF THE FACILITIES AND FABRICATION TECHNOLOGIES OF REFLECTIVE LAMINATES COUPLED WITH THE COMPANY'S RECOGNIZED EXPERIENCE PROVIDES THIS CAPABILITY.

REFLECTIVE LAMINATES MANAGEMENT ENGINEERS AND TECHNICIANS HAVE ESTABLISHED A UNIQUE CAPABILITY IN PLASTIC FABRICATION. THE LAMINATION PROCESS OF BONDING LAYERS OF MATERIAL INTO A SHAPE IS AN ACCURATELY CONTROLLED FABRICATION SYSTEM. FABRIC, IMPREGNATED WITH RESIN IS FORMED INTO A DESIRED SHAPE, THEN CONVERTED INTO A DENSE TOUGH HOMOGENOUS SOLID BY SIMULTANEOUSLY APPLYING HEAT AND PRESSURE.

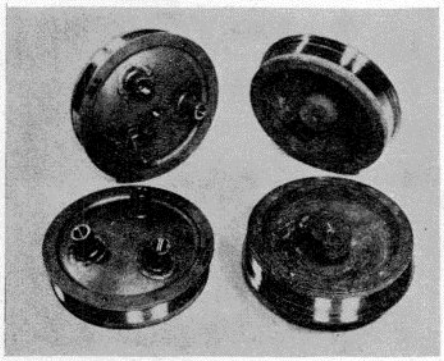
REFLECTIVE LAMINATES CURRENTLY FURNISHES SOPHISTICATED STRUCTURES FOR MANY AIRCRAFT, MISSILES AND SPACECRAFT APPLICATIONS MADE FROM THE FOLLOWING MATERIALS; "E" GLASS NYLON, SILICA, CARBON, GRAPHITE, AND ASBESTOS IMPREGNATED WITH POLYESTER, EPOXY, PHENOLIC, SILICONE RUBBER OR MANY OTHER RESIN FORMULATIONS.



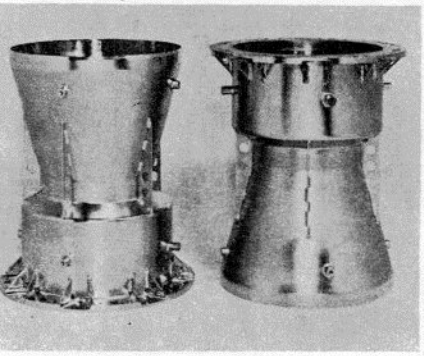
Silica-Phenolic Cover Compression Molded



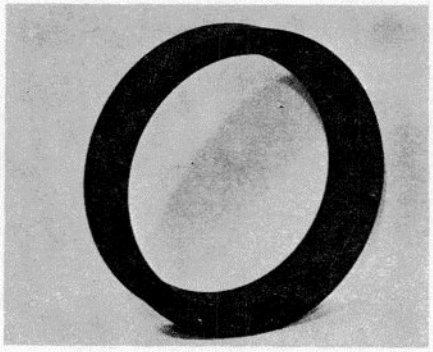
Nylon Fabric - Phenolic Radome Silver Metalized



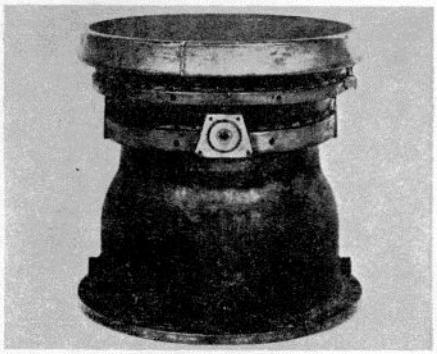
Titanium Closures With Molded Silica Insulation



tanium Ablative Thrust Chamber



Carbon Phenolic Fabric Tape Wound Nozzle Insulator



Maraging Steel Thrust Chamber With Silica Tape Insulation